

Threshold Strategy & Alert Prioritisation Memo

Explainable Fraud Intelligence Framework

JP Morgan Data Analytics Virtual Experience (Forage)

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1. Executive Summary

Fraud detection thresholds directly influence the balance between fraud capture, false positives and investigator workload. This memo documents the threshold strategy used within the Explainable Fraud Intelligence Framework and explains how model probabilities are translated into operational alert prioritisation.

Logistic Regression is retained as the selected operational model because it provides a strong balance between fraud detection sensitivity, transparency and investigator usability. XGBoost is retained as a benchmark model and for SHAP-based explainability rather than as the operational decision model.

The recommended operating point is the balanced threshold of 0.50, supported by risk-based prioritisation, reason codes, recommended investigator actions, SLA categories and operational KPI monitoring. The threshold strategy is intended to support human-led investigation decisions, not automated fraud rejection.

2. Threshold Evaluation

Three operating thresholds were evaluated to understand the trade-off between detection coverage and operational workload. The values below reflect the final notebook threshold evaluation and should be interpreted as decision-support evidence for investigator queue design.

Operating Option	Threshold	Alerts Flagged	Precision	Recall	False Positive Rate	Best For
Conservative	0.70	95	0.084	0.200	0.091	Workload containment
Balanced	0.50	263	0.080	0.525	0.252	Recommended operating point
Aggressive	0.30	547	0.049	0.675	0.542	Maximum fraud capture

Table 1. Final threshold evaluation summary for operational fraud alert prioritisation.

3. Operating Options

3.1 Conservative Threshold - 0.70

The conservative threshold limits the investigation queue and reduces false positive volume. This option is suitable when investigator capacity is highly constrained, but it increases missed fraud exposure because recall is lower.

3.2 Balanced Threshold - 0.50

The balanced threshold provides the preferred operating point for this project. It improves fraud recall compared with the conservative option while keeping the investigation queue more manageable than the aggressive option. This threshold is recommended as the baseline for operational review because it balances fraud detection effectiveness with investigator workload.

3.3 Aggressive Threshold - 0.30

The aggressive threshold increases fraud capture and reduces missed fraud, but it also generates a substantially larger investigation queue and a higher false positive rate. This option is useful when maximum fraud capture is prioritised over workload containment.

4. Alert Prioritisation Framework

Threshold selection should not operate in isolation. The framework combines model probability outputs with operational risk categories to direct investigator effort towards the highest-risk transactions. Risk categories, reason codes and recommended actions provide additional context so analysts understand why a transaction was prioritised.

Risk Tier	Operational Action	Investigation Priority
High Risk	Immediate escalation	Same Day Review
Medium Risk	Analyst review	Review Within 24 Hours
Low Risk	Routine monitoring	Weekly Monitoring

Table 2. Operational alert prioritisation structure used within the framework.

5. Operational KPI Monitoring

Threshold performance should be monitored using operational KPIs rather than model accuracy alone. The final dashboard and notebook framework include alert quality, detection coverage, investigation waste exposure, missed fraud exposure, active investigation queue and queue risk severity. These KPIs help fraud operations teams understand whether the selected threshold remains aligned with business risk appetite and investigation capacity.

6. Governance Considerations

Any threshold adjustment should be documented and reviewed through a governance process. Recommended controls include periodic threshold review, false-positive monitoring, recall monitoring, SLA adherence tracking, investigator feedback review and model drift assessment before production deployment. The operational alert flag was excluded from model training to reduce leakage risk. Final decisions remain subject to human investigator review, supporting responsible AI principles and ensuring that the framework remains a decision-support capability rather than an automated enforcement mechanism.

7. Recommendation

The recommended baseline operating threshold is 0.50. This balanced threshold provides a practical starting point for the simulated fraud operations environment by improving fraud capture while maintaining a manageable investigation workload. For production use, the threshold should be recalibrated regularly using live fraud outcomes, investigation capacity, business risk appetite and analyst feedback.

8. Clarifying Note

The alert volumes in the threshold evaluation table represent model-threshold flagged cases. These are separate from the dashboard high-risk transaction count, which reflects the operational risk-category framework. Both views support the same objective: prioritising fraud investigation activity using explainable, risk-based decision support.